When a permutation action detects abstract isomorphism

Human Compression project: L071 recovery

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Let G be a finite group and let $\pi : G \hookrightarrow \mathrm{Sym}(\Omega)$ be a faithful permutation representation. An ordered generating tuple $t = (t_0, \ldots, t_{r-1})$ of involutions defines a graph on Ω: join x to $\pi(t_i)x$ with an edge of colour i whenever these vertices are distinct. Colours are retained, including parallel edges with different colours. A fixed point of $\pi(t_i)$ is exactly a vertex with no edge of colour i incident to it. Thus the graph determines every permutation $\pi(t_i)$, including its fixed points.

This note supplies the precise action hypothesis in Leemans–Mixer, *Algorithms for classifying regular polytopes with a fixed automorphism group*, Theorem 4.1. The proof uses the elementary classification of finite G-sets by their point stabilizers. That classification, and the normalizer action below, are standard ideas; no new general group-theoretic result is claimed.

Theorem 1. *For two ordered generating involution tuples t, t' of G, their coloured graphs are isomorphic if and only if some element of $\mathrm{N}_{\mathrm{Sym}(\Omega)}(\pi(G))$ simultaneously conjugates $\pi(t_i)$ to $\pi(t'_i)$ for every i. The automorphisms of G induced in this way form the image of the homomorphism*

$$\mathrm{N}_{\mathrm{Sym}(\Omega)}(\pi(G)) \longrightarrow \mathrm{Aut}(G), \qquad f \longmapsto \big[g \mapsto \pi^{-1}(f\pi(g)f^{-1})\big],$$

whose kernel is $\mathrm{C}_{\mathrm{Sym}(\Omega)}(\pi(G))$.

Write the orbit decomposition of Ω as

$$\Omega \cong \coprod_{[H]} m_{[H]}\, G/H,$$

where $[H]$ ranges over conjugacy classes of subgroups of G and $m_{[H]}$ is a nonnegative integer. An automorphism α belongs to the displayed image if and only if

$$m_{[H]} = m_{[\alpha(H)]} \quad \text{for every } H. \tag{1}$$

In particular, if these multiplicities are invariant under every abstract automorphism, coloured graph isomorphism is equivalent to abstract equivalence of the generating tuples under $\mathrm{Aut}(G)$.

Proof. A colour-preserving graph isomorphism f sends the unique neighbour of x of colour i, when present, to the corresponding neighbour of $f(x)$; it also sends vertices with no such neighbour to vertices with no such neighbour. Consequently $f\pi(t_i)f^{-1} = \pi(t'_i)$ on every vertex. Since both tuples generate G, this identity gives $f\pi(G)f^{-1} = \pi(G)$. Conversely, simultaneous conjugation of these permutations preserves exactly the edges and colours of the graphs. The assertion about the homomorphism and its kernel follows directly from its formula and the faithfulness of π.

Suppose f induces α. The relation $f(gx) = \alpha(g)f(x)$ shows that the stabilizer of $f(x)$ is $\alpha(G_x)$. Thus f permutes the transitive orbits according to $[H] \mapsto [\alpha(H)]$, proving the necessity of (1). For sufficiency, match the copies of G/H bijectively with the copies of $G/\alpha(H)$ using (1), and on each matched pair use

$$gH \longmapsto \alpha(g)\alpha(H).$$

This is well defined and bijective, and has the required equivariance. If the chosen representative of the target stabilizer class differs from $\alpha(H)$, compose with the usual equivariant coset isomorphism for conjugate subgroups. The maps on the disjoint orbits assemble into a permutation f inducing α. Finally abstract equivalence of tuples is exactly simultaneous application of an automorphism to their entries; surjectivity of the displayed homomorphism proves the last assertion. □

Corollary 2. *For every finite G, its regular action detects abstract isomorphism of all ordered generating involution tuples by coloured graph isomorphism. More generally, any faithful finite G-set with an $\mathrm{Aut}(G)$-invariant multiset of stabilizer conjugacy classes does so. For string C-group tuples, the same statement detects isomorphism of the associated regular abstract polytopes. Reversing the colour order detects duality.*

Proof. The regular action has one orbit with trivial stabilizer. Every automorphism preserves that stabilizer class, so Theorem 1 applies. Explicitly, on the underlying set G, the permutation $f_\alpha : g \mapsto \alpha(g)$ conjugates each left multiplication L_h to $L_{\alpha(h)}$. The more general assertion is the last part of the theorem. The marked-group correspondence for regular abstract polytopes identifies their isomorphisms with automorphisms carrying the ordered distinguished generators to one another; that standard correspondence is retained from the common theory. Reversal of the ordered generators is the definition of the dual marked polytope, giving the final assertion. □

Why two unrelated embeddings do not suffice. Take $G = \langle a, b\rangle \leq \mathrm{Sym}(6)$ with $a = (1\ 2)$ and $b = (3\ 4)(5\ 6)$. This is a Klein four group. Both (a, b) and (b, a) are string C-group tuples: their distinct order-two subgroups meet trivially, which verifies every intersection of subgroups generated by subsets. They are abstractly equivalent under the automorphism interchanging a and b. Nevertheless their coloured graphs are not isomorphic, since colour zero has one edge in the first graph and two in the second. The abstract group $\mathrm{Aut}(G) \cong \mathrm{Sym}(3)$ also embeds in $\mathrm{Sym}(6)$. Thus merely embedding both abstract groups in the same symmetric group does not supply the needed compatibility. In this example the normalizer cannot induce the automorphism interchanging a and b, because conjugation preserves cycle type.

The source's proof uses the stronger assumption that automorphisms act on the embedded G by permutation conjugation. Under that intended hypothesis, Theorem 1 proves its graph criterion. Its literal statement requires this qualification. In the reverse implication a normalizing permutation induces an automorphism; it need not itself belong to a separately embedded copy of $\mathrm{Aut}(G)$. The centralizer kernel in Theorem 1 makes this distinction explicit.

Use in the finite classifications. The natural action of A_n for $n \geq 7$ satisfies the criterion, using the retained classical theorem that all its automorphisms are induced by S_n. In other models it is enough to exhibit a subgroup $A \leq \mathrm{Sym}(\Omega)$ normalizing $\pi(G)$ and verify that $|A|/|\,\mathrm{C}_A(\pi(G))| = |\,\mathrm{Aut}(G)|$. The image is then the full automorphism group. This is the exact guard used by the accompanying enumerations. The regular action is always available, while the stabilizer-multiplicity criterion explains when a smaller action suffices.

This correction and elementary reproof do not settle an open conjecture. The numerical classifications and their full finite outputs remain separate premises of the L071 recovery. This note alone supplies no whole-original, formalization or compression credit.